Week 12 (CS 418 @ UIC)

Week 12 Slide Deck

Intro to Network Analysis

Jack Bandy 2026

Topic Title Placeholder

CS 418 · Week 12 · 🟠 Western 🟠


STOP
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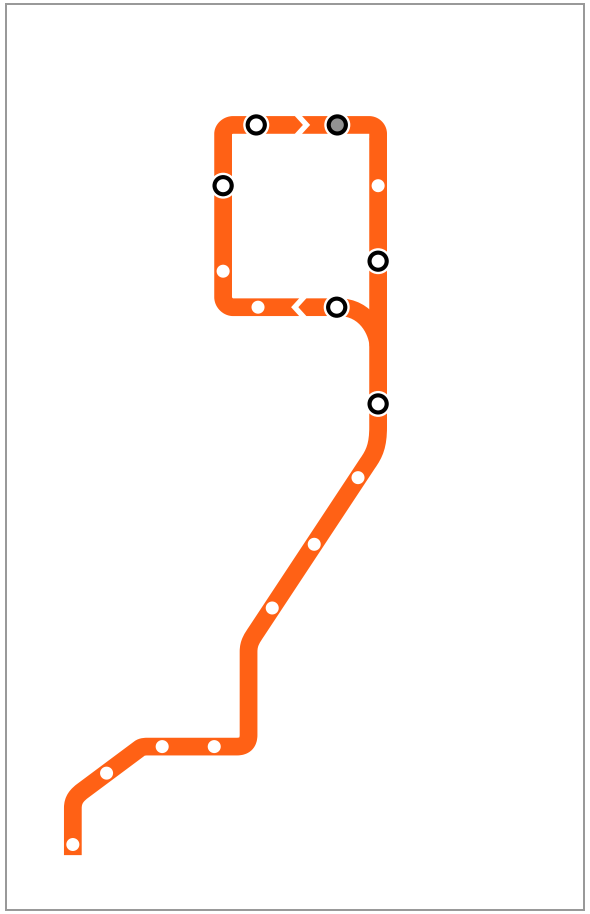
Western
NO PARKING ANY TIME

What is a Graph?

A graph represents objects and relationships

- $G = (V, E)$
- V = vertices (objects)
- E = edges (relationships)

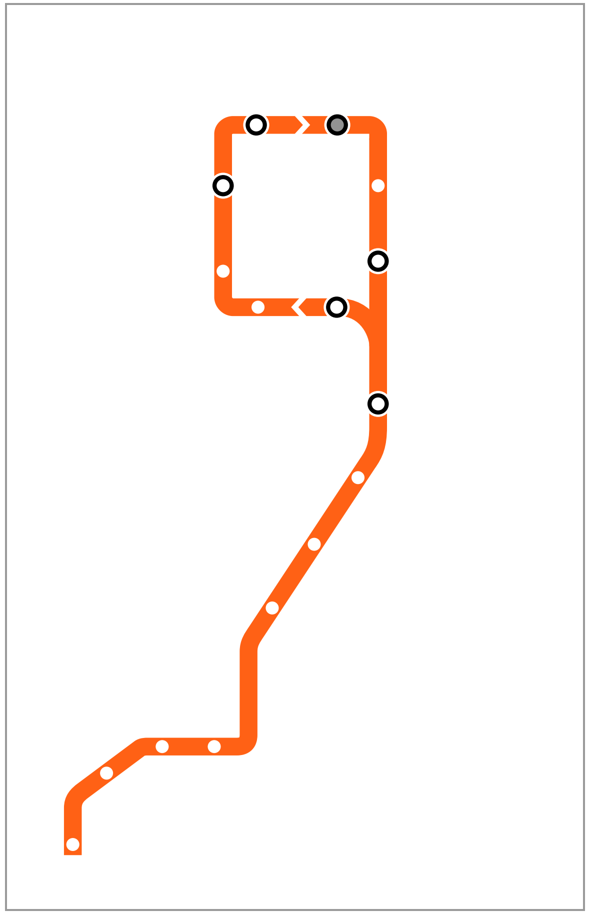


Social network graph *(placeholder)*

Graph Applications

Graphs model many real-world systems:

- World Wide Web
- Social networks
- Transportation networks
- Computer networks

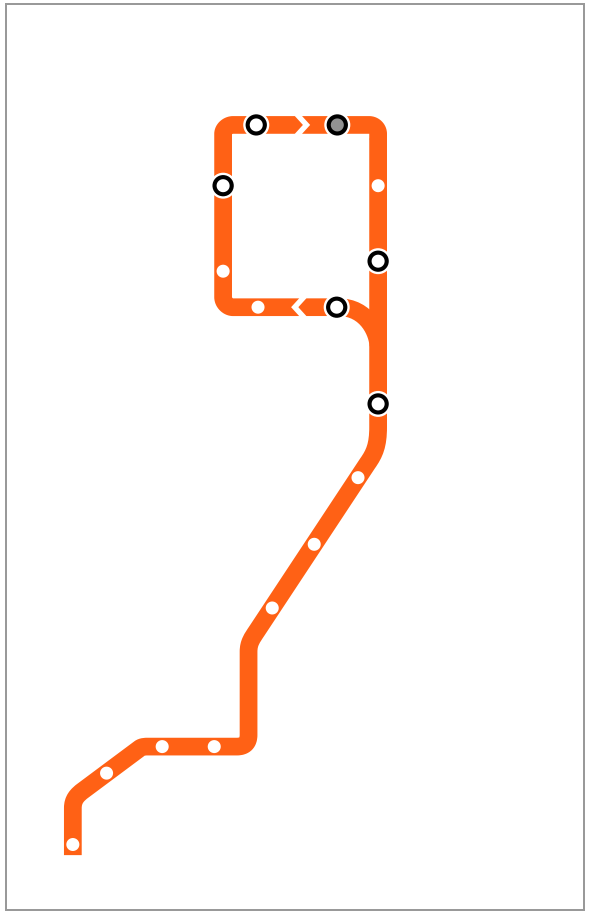


Graph applications *(placeholder)*

Graph Terminology

Key concepts:

- **Endpoints:** Vertices connected by an edge
- **Adjacent:** Vertices connected by an edge
- **Degree:** Number of edges on a vertex
- **Self-loop:** Edge from vertex to itself

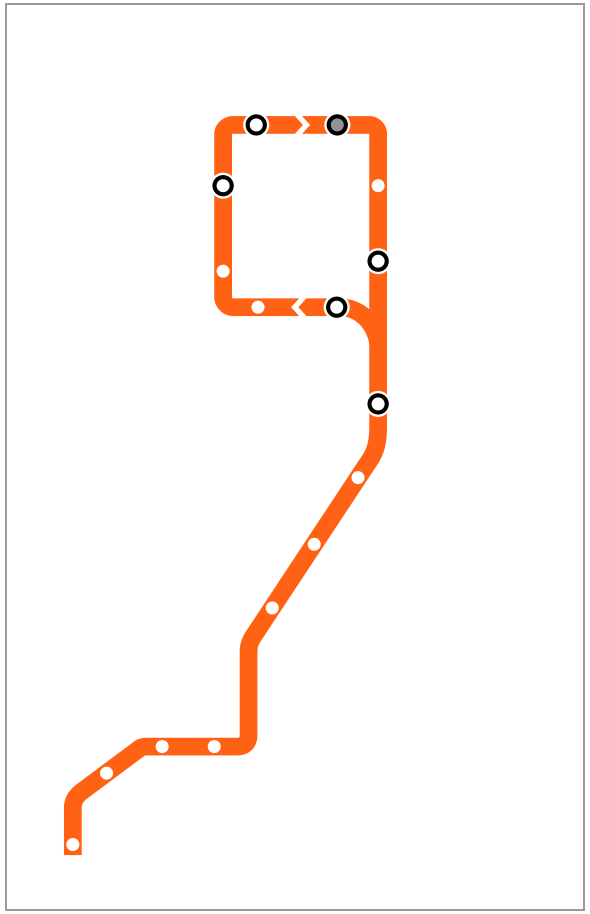


Graph terminology *(placeholder)*

Adjacency Matrix

Matrix representation of graphs

- Efficient for dense graphs
- Easy to check connections

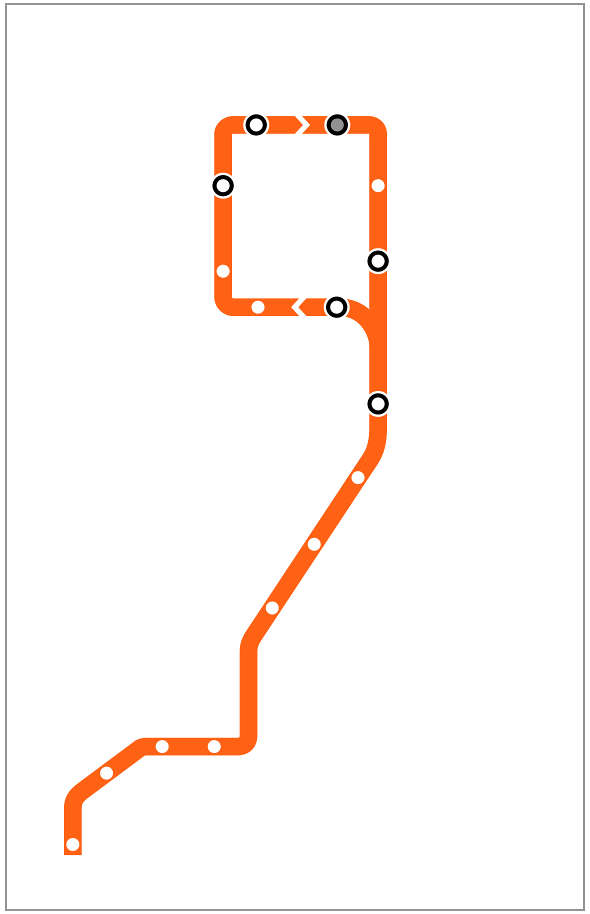


Adjacency matrix example *(placeholder)*

Types of Graphs

Three important types:

- **Directed:** Edges have direction
- **Weighted:** Edges have values
- **Attributed:** Nodes have properties

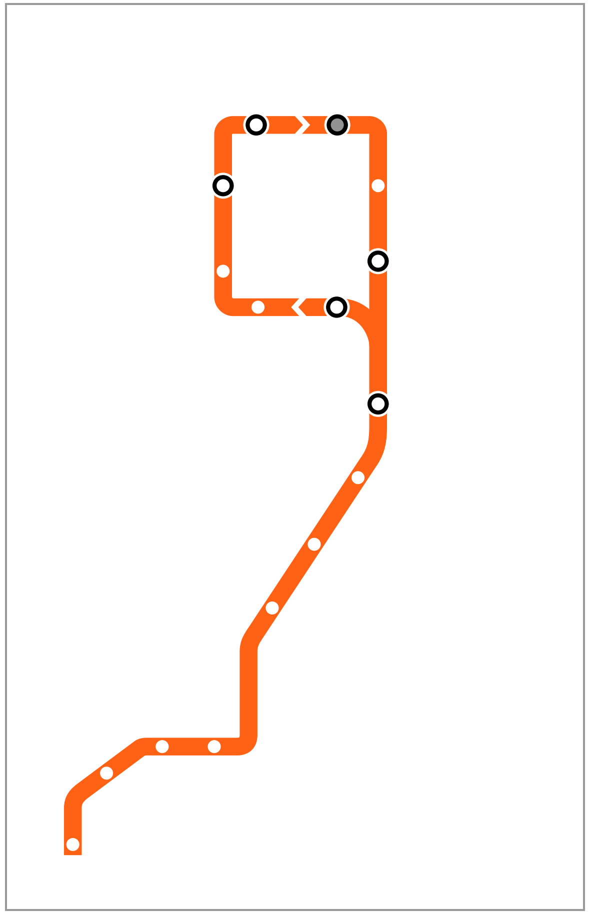


Graph types *(placeholder)*

Community Detection

Finding groups in networks

- Communities = densely connected groups
- Sparse connections between groups
- Social groups in social networks



Community detection *(placeholder)*

Goodness Metrics

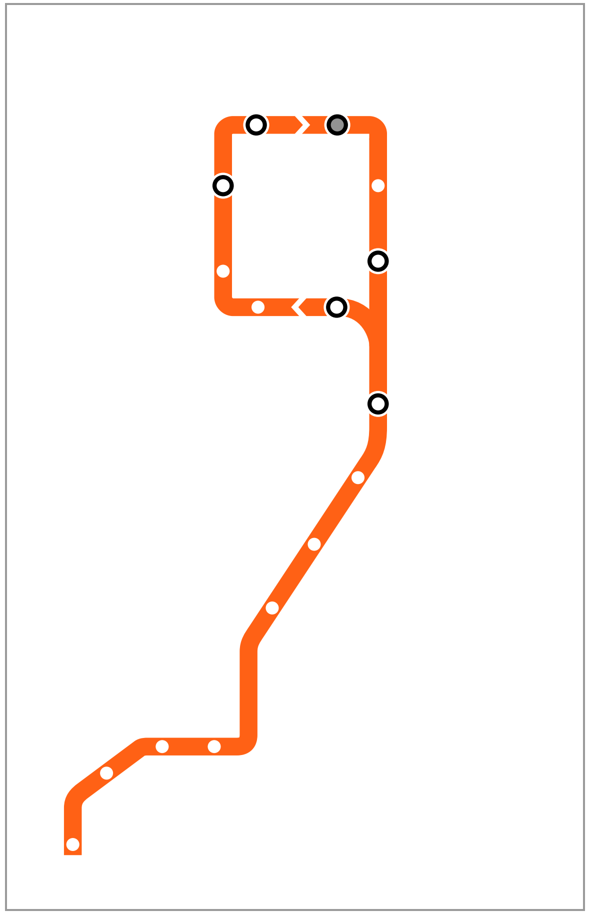
Measuring community quality

Density: Ratio of actual to possible edges

$$\frac{2E_s}{S(S-1)}$$

Conductance: Fraction of edges leaving

$$\frac{O_s}{2E_s + O_s}$$

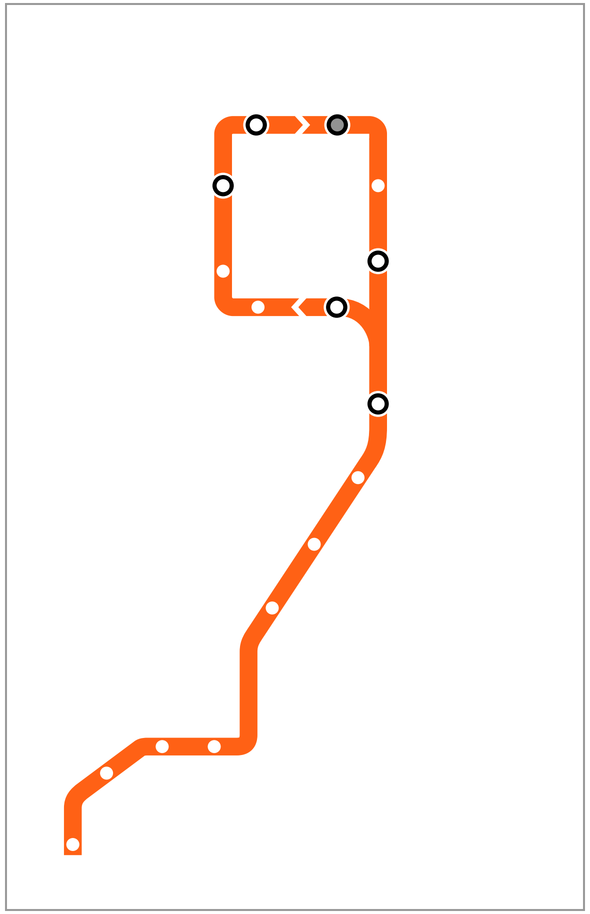


Modularity concept *(placeholder)*

Modularity

Most popular community detection metric

- Actual edges – Expected edges
- Higher Q = better communities
- Maximizing Q is NP-complete

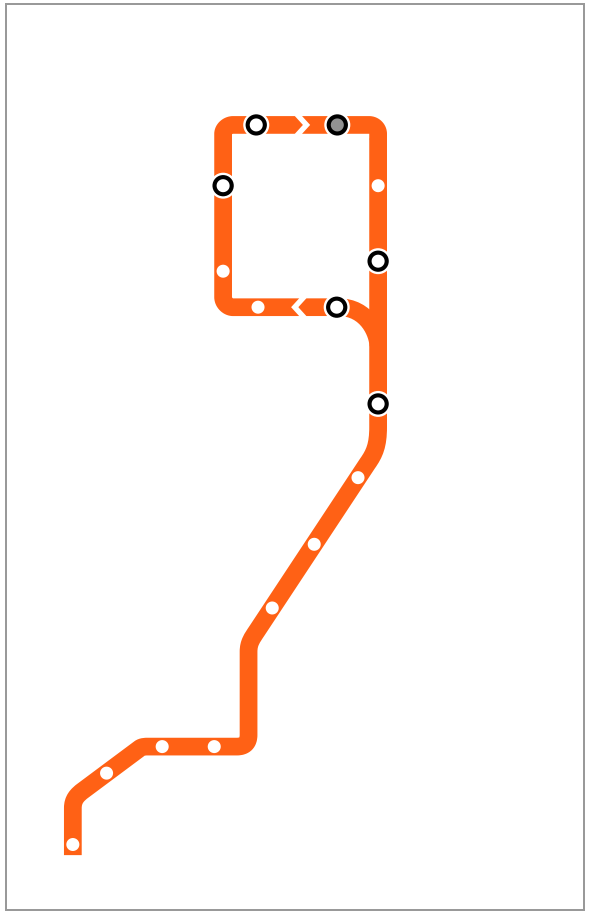


Modularity visualization *(placeholder)*

Louvain Method

Greedy modularity optimization

1. Start with each node as community
2. Move nodes to maximize Q
3. Aggregate communities into supernodes
4. Repeat until convergence



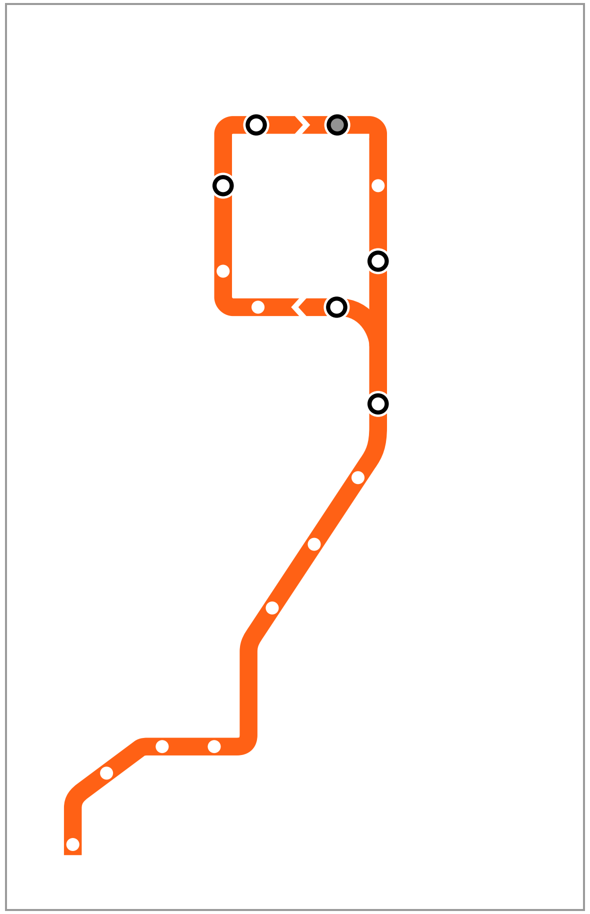
Community detection *(placeholder)*

Network Diffusion

How events spread through networks

Key questions:

- Will it spread?
- Which nodes to target?
- How to stop it?

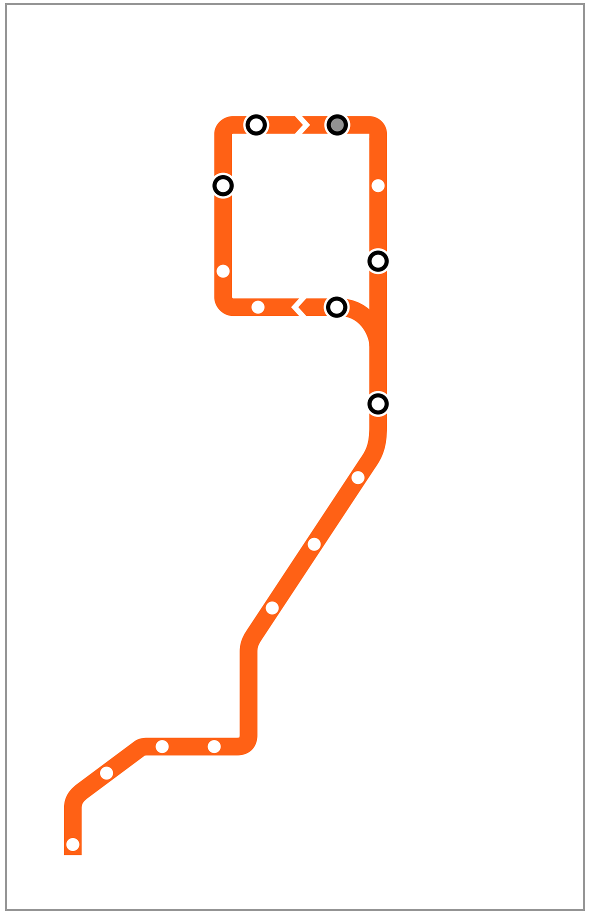


Simple network *(placeholder)*

Broadcast vs. Viral

Two ways content reaches a crowd
(Goel et al. 2016)

- **Broadcast:** one source reaches many directly (e.g. a front page, the Super Bowl)
- **Viral:** person-to-person, over many generations
- Real cascades mix the two



Broadcast (shallow) vs. viral (deep) diffusion trees *(placeholder)*

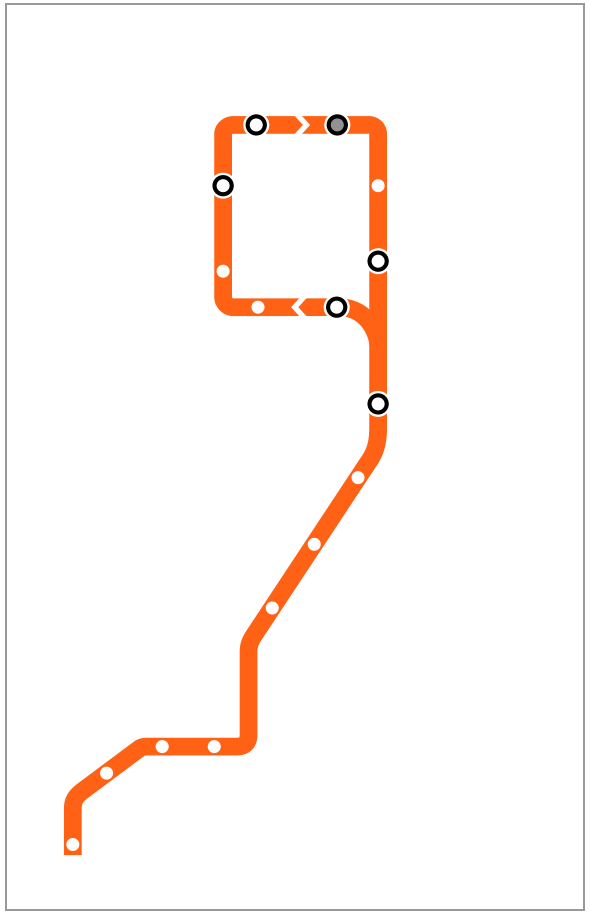
Structural Virality

A single number for tree shape (Goel et al. 2016)

Average distance between all pairs of nodes in a diffusion tree (the *Wiener index*):

$$\nu(T) = \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1}^{n} d_{ij}$$

- $\nu(T) \approx 2$: pure broadcast (a star)
- Large $\nu(T)$: deep, multigenerational spread

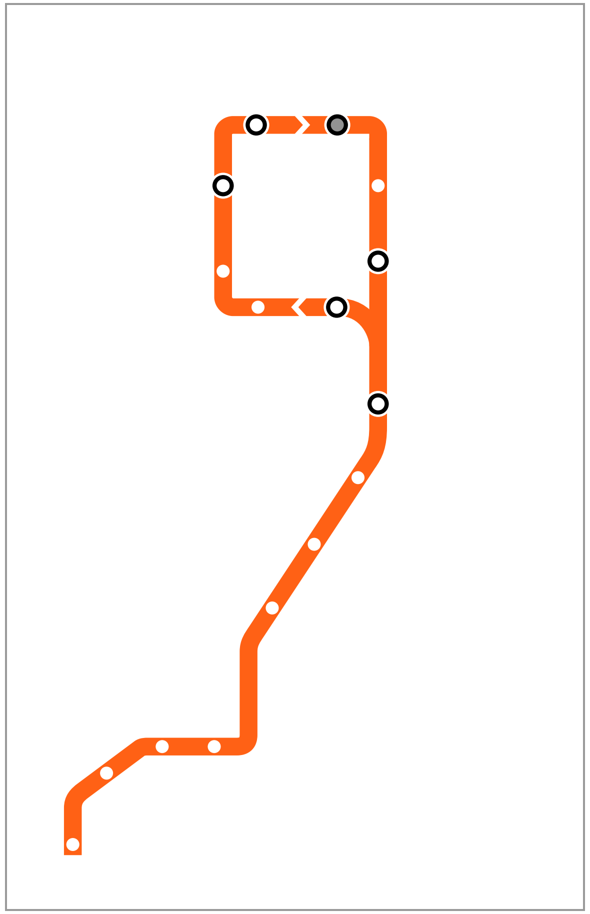


Diffusion tree with pairwise distances *(placeholder)*

What Spreads Online?

~1 billion Twitter diffusion events (Goel et al. 2016)

- Over 99% of cascades die in one generation
- Virality varies widely (**no** "tipping point")
- Popularity is driven mostly by the largest broadcast

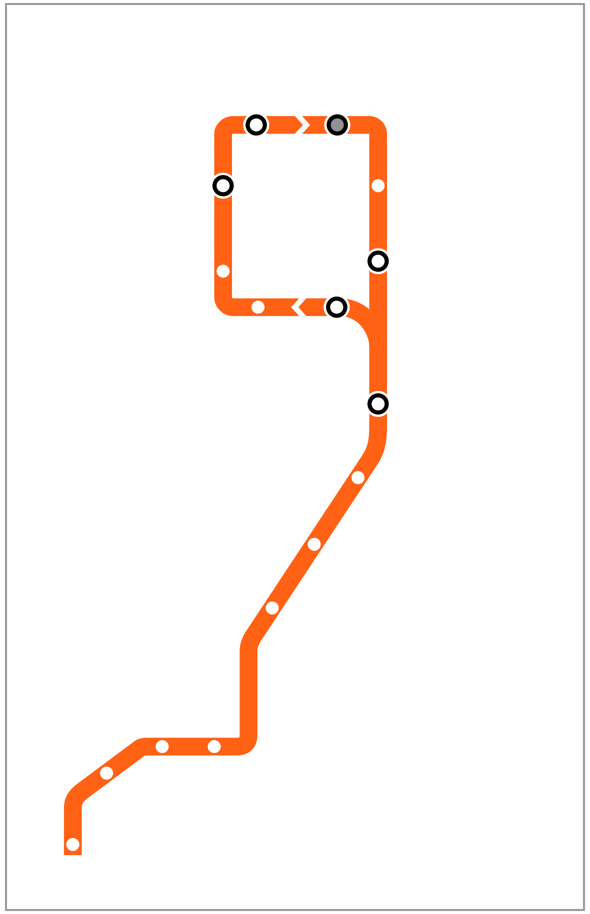


Distribution of cascade sizes and structural virality *(placeholder)*

Virus Propagation Models

VPMs simplify disease spread

- How virulent?
- Recovery rate?
- Immunity?
- Immunity duration?

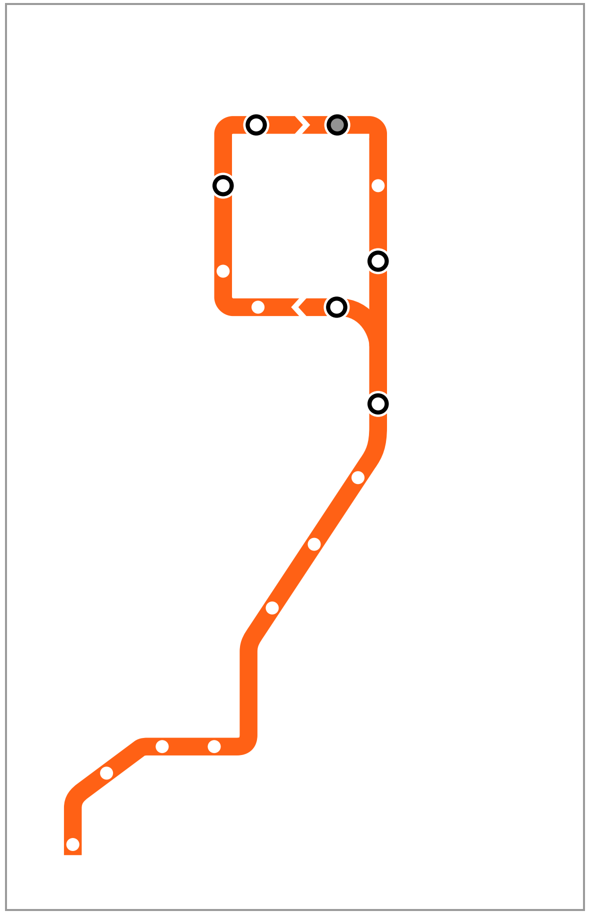


SIS model *(placeholder)*

SIS Model

Susceptible → Infected → Susceptible

- Example: flu
- β = transmission probability
- δ = healing probability
- No immunity

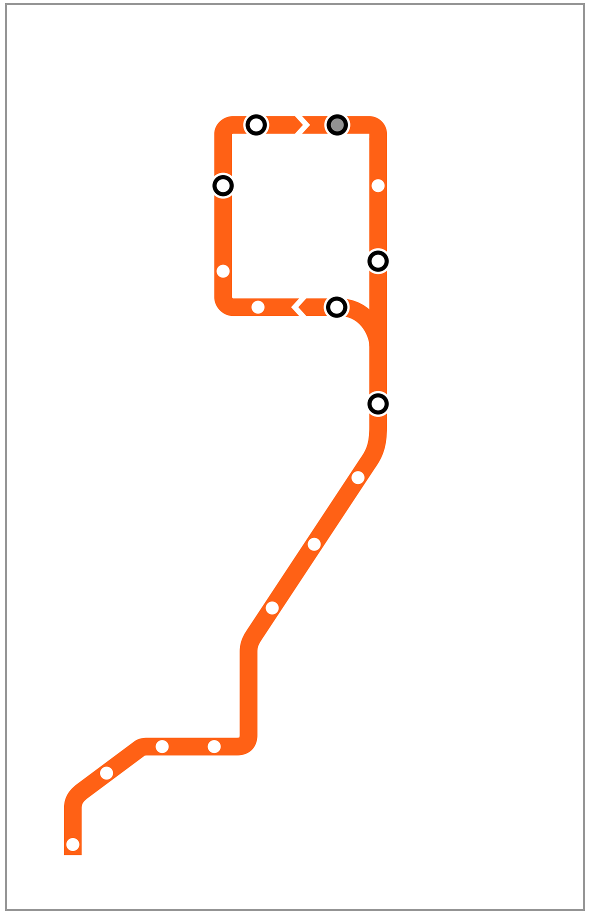


SIS model diagram *(placeholder)*

SIR Model

Susceptible → Infected → Recovered

- Example: mumps
- β = transmission probability
- δ = healing probability
- Life-time immunity

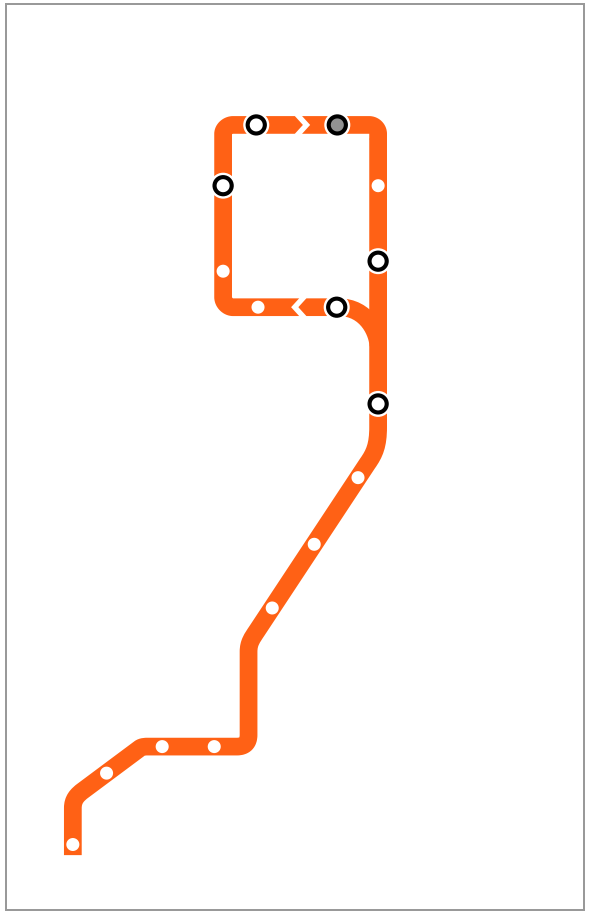


SIR model diagram *(placeholder)*

$SIRS$ Model

Temporary immunity

- Example: whooping cough
- γ = immunity loss probability
- Immunity wears off over time

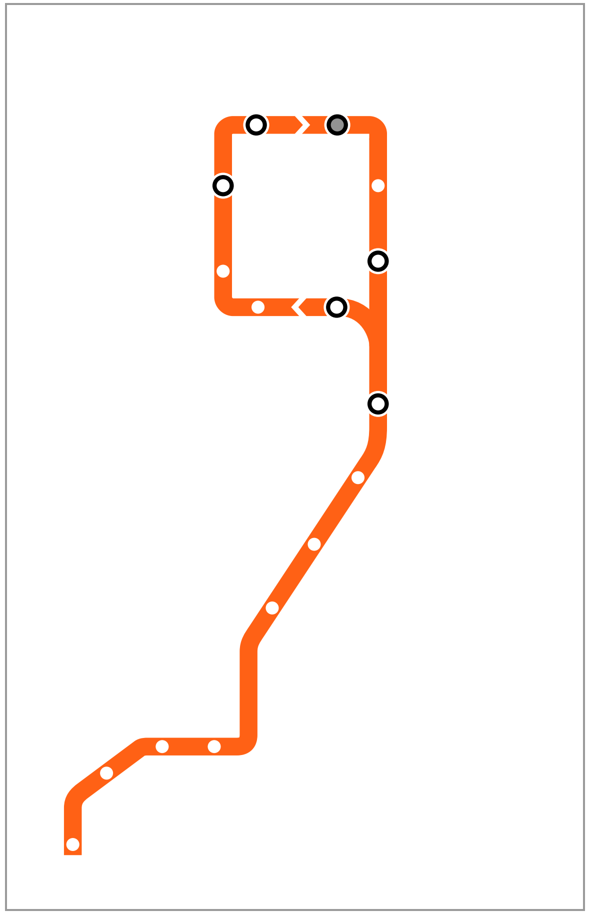


SIRS model diagram *(placeholder)*

$SEIV$ Model

Most complex VPM

- **E** = Exposed (incubation)
- **V** = Vaccinated
- ε = virus maturation
- θ = vaccination rate



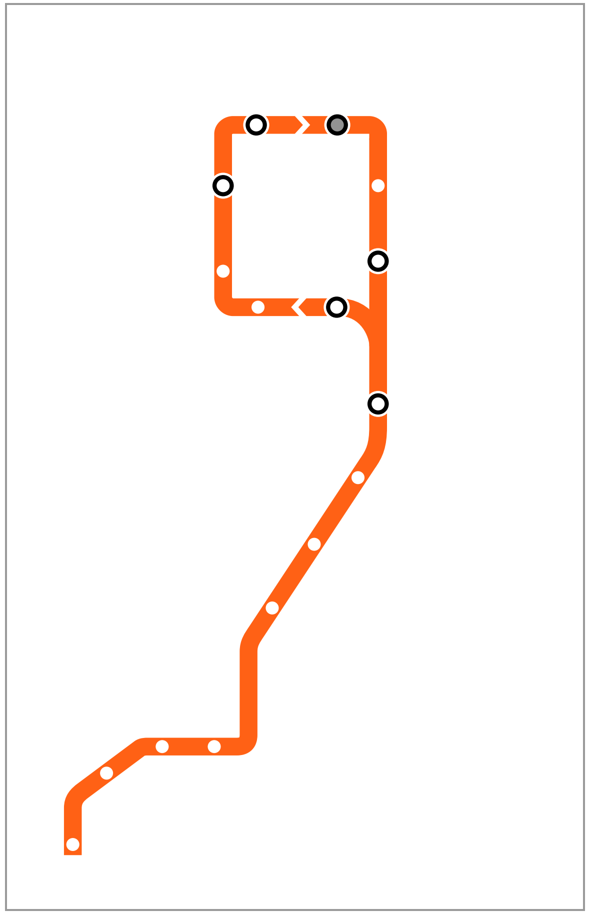
SEIV model diagram *(placeholder)*

Epidemic Threshold

Critical point for virus spread

$$s = \lambda_1 \cdot C_{VPM}$$

- $s < 1$: Virus dies out
- $s > 1$: Epidemic spreads
- λ_1 = network connectivity

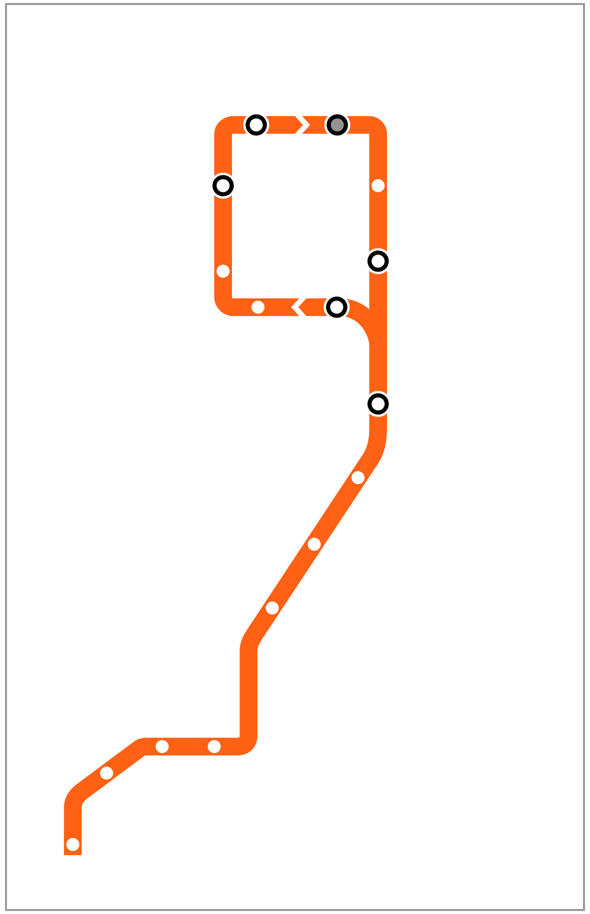


Epidemic threshold *(placeholder)*

Network Connectivity

Why spectral radius matters

- Average degree ignores paths
- λ_1 considers all path lengths
- Better predicts spread



Network connectivity examples
(placeholder)

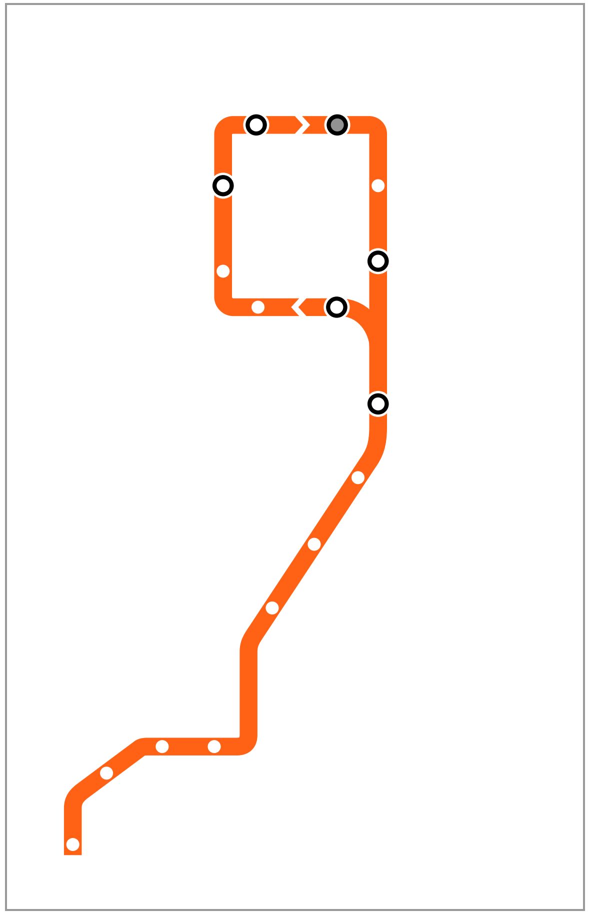
Effective Strength

Predicting epidemic spread

$$s = \lambda_1 \cdot C_{VPM}$$

For SIS: $s = \lambda_1 \cdot \frac{\beta}{\delta}$

- Higher connectivity → more spread
- Higher transmission → more spread
- Higher healing → less spread



Epidemic threshold visualization *(placeholder)*

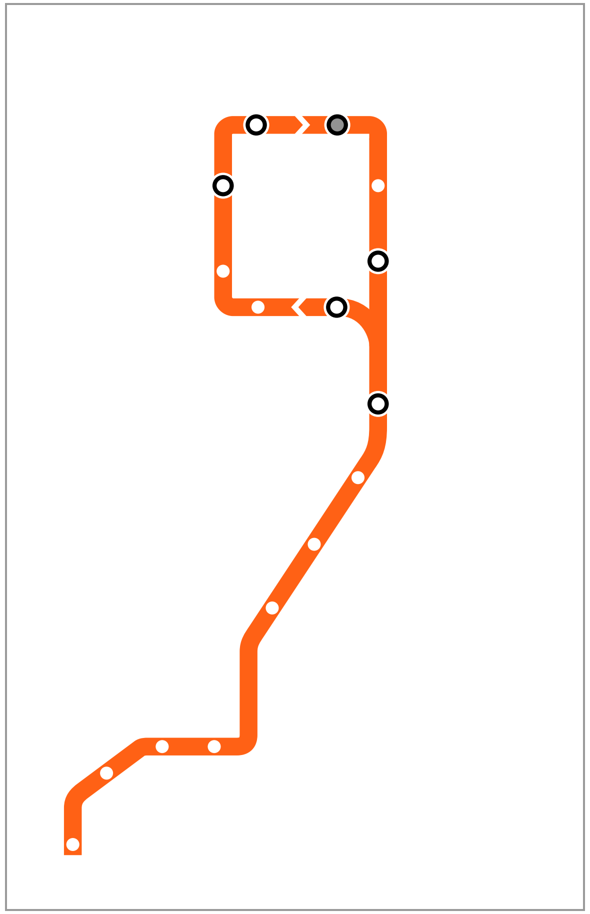
Immunization Problem

Which k nodes to vaccinate?

Goal: Minimize λ_1

Approaches:

- Random (poor)
- High-degree nodes (better)
- NetShield (best)

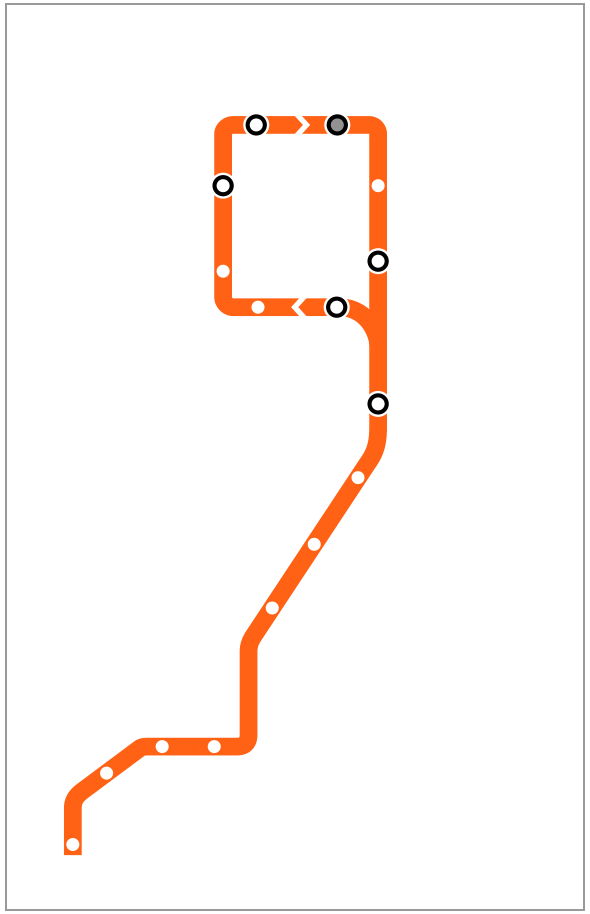


Network connectivity *(placeholder)*

NetShield Algorithm

Efficient immunization

- Approximates eigen-drop
- Greedy selection
- Outperforms heuristics



Community detection showing key nodes *(placeholder)*

References & Credits

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GitHub source: https://github.com/jackbandy/data-science-fun/blob/main/docs/slides/week12.md.

Last modified and compiled June 23, 12:35h Central (Chicago) time.

Slides developed using materials from Elena Zheleva and Gonzalo Bello Lander, the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart. Network analysis content heavily adapted from Gonzalo Bello's **CS 418 - Lecture Slides 20 - Network Analysis.pdf**.

Slide deck built with Quarto revealjs.

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